

NEXOMATE — Technical Architecture Document

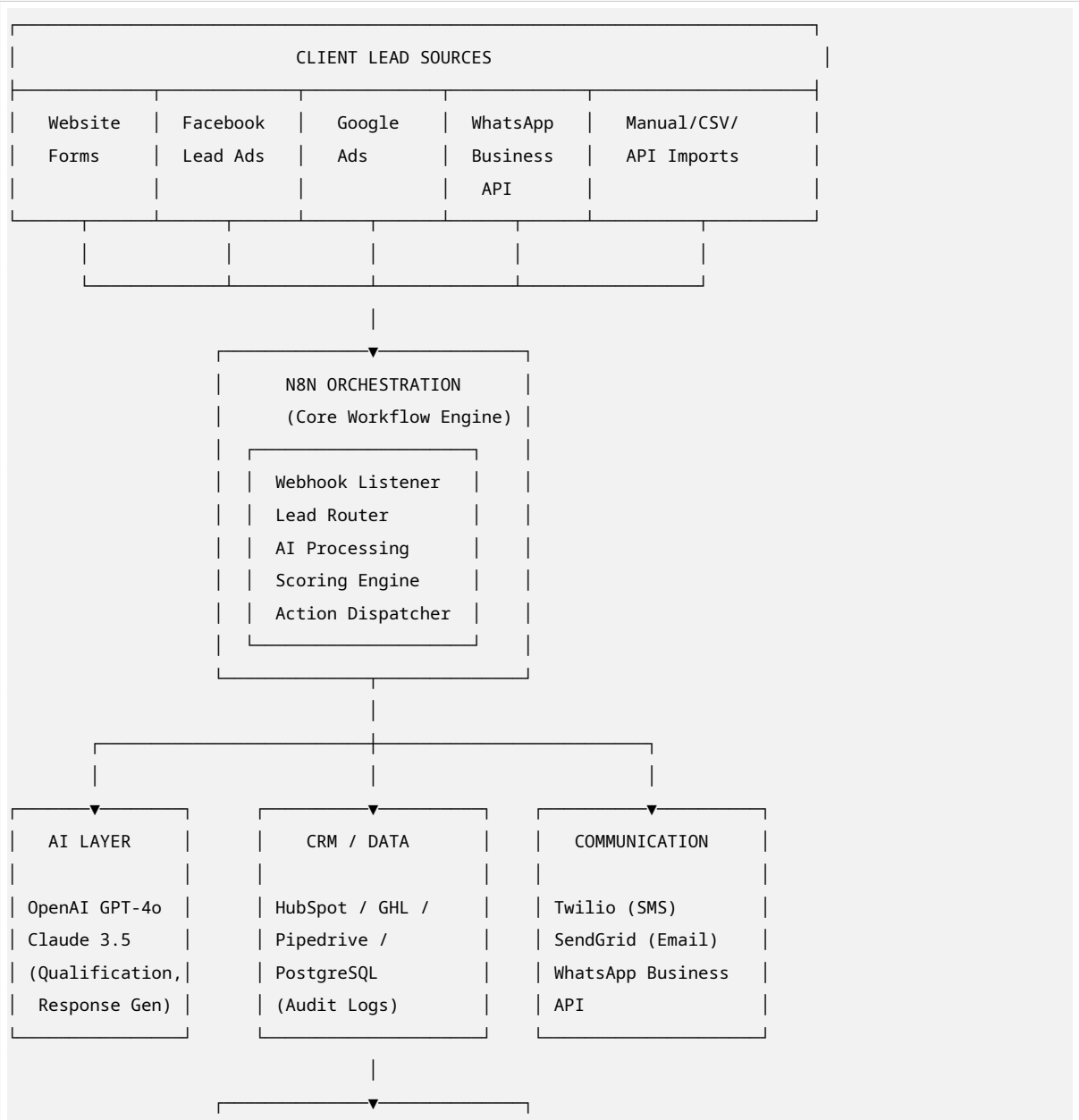
AI Lead Conversion System | v1.0 | Prepared for Founder’s Review

1. EXECUTIVE SUMMARY

This document outlines the technical architecture for Nexomate’s AI-powered lead conversion platform. The system is designed to capture, qualify, score, and nurture inbound leads for solar installation and interior design companies across international markets.

Core Principle: Build once as a template, deploy per client with configuration — not custom code.

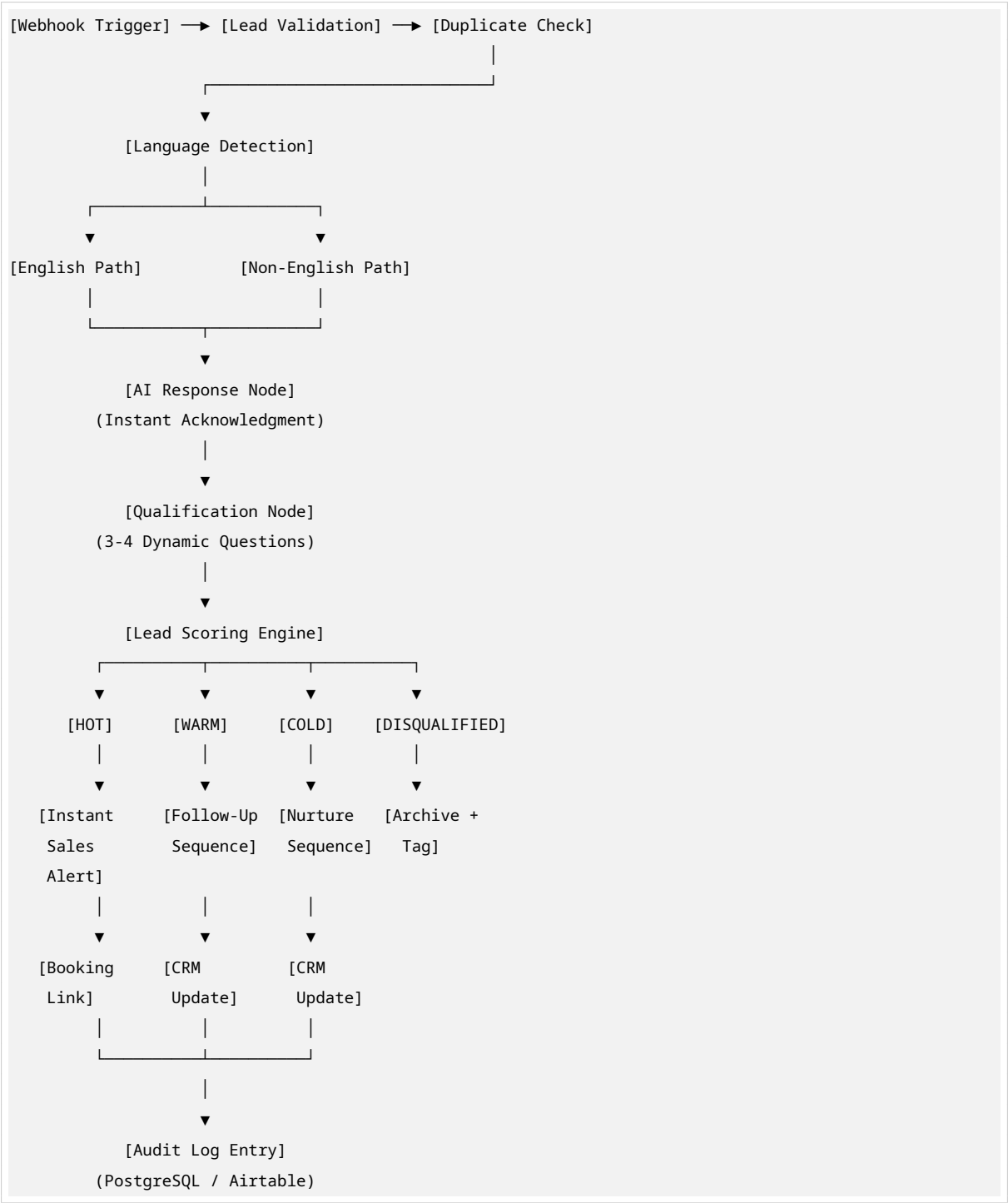
2. SYSTEM ARCHITECTURE (High-Level)



	SCHEDULING & APPOINTMENT Calendly / SavvyCal	
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3. N8N WORKFLOW ARCHITECTURE

3.1 Master Lead Processing Pipeline



3.2 Follow-Up Sequence Engine (WARM Leads)

Day 0 (T+0 min):	AI Instant Response + Qualification
Day 0 (T+2 hrs):	Value-driven follow-up (if no reply)
Day 1 (T+24 hrs):	Case study / testimonial share
Day 3 (T+72 hrs):	Direct booking link + urgency
Day 7 (T+168 hrs):	Final attempt + long-term nurture tag

3.3 Nurture Sequence Engine (COLD Leads)

Week 1:	Educational content (solar ROI calculator, design trends)
Week 2:	Social proof (recent installations, client reviews)
Week 3:	Soft re-qualification ("Still considering solar?")
Week 4+:	Monthly newsletter automation

4. TECHNOLOGY STACK

Layer	Primary Tool	Alternative	Justification
Workflow Engine	n8n (Self-hosted)	Make.com, Zapier	Full control, no per-task limits, custom nodes
AI/LLM	OpenAI GPT-4o	Claude 3.5 Sonnet, Gemini 1.5	Best cost/performance for structured output
CRM (Primary)	GoHighLevel (GHL)	HubSpot, Pipedrive	Built-in SMS/email, popular in US solar market
CRM (Enterprise)	HubSpot	Salesforce	Better for larger international clients
Database	PostgreSQL (Supabase)	Airtable, MongoDB	Structured data, audit compliance, low cost
Email	SendGrid	Mailgun, AWS SES	Deliverability, templates, analytics
SMS	Twilio	MessageBird, Vonage	Global coverage, programmable
WhatsApp	WhatsApp Business API	360dialog, WATI	Required for EU, LATAM, India, Middle East
Scheduling	SavvyCal	Calendly, Acuity	Better timezone handling, white-label
Voice AI	Bland AI / Retell	Vapi, Synthflow	Premium upsell —3-5x conversion boost
Hosting	DigitalOcean / Hetzner	AWS, GCP	Cost-effective for early stage
Monitoring	n8n Error Workflows + Slack	Sentry, Datadog	Immediate failure alerts
Secrets Mgmt	n8n Built-in + .env	HashiCorp Vault	Sufficient for current scale

5. DATA MODEL

5.1 Core Lead Entity

```
CREATE TABLE leads (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  client_id UUID NOT NULL,  
  source VARCHAR(50) NOT NULL,           -- website, facebook, google, whatsapp  
  source_detail VARCHAR(255),           -- specific campaign, page URL  
  
  -- Contact Info  
  first_name VARCHAR(100),  
  last_name VARCHAR(100),  
  email VARCHAR(255),  
  phone VARCHAR(50),  
  country_code VARCHAR(5),             -- +1, +44, +91, etc.  
  
  -- Qualification Data (JSONB for flexibility)  
  qualification_answers JSONB,  
  
  -- Scoring  
  lead_score VARCHAR(20) CHECK (lead_score IN ('HOT', 'WARM', 'COLD', 'DISQUALIFIED')),  
  score_reason TEXT,  
  
  -- Status Tracking  
  status VARCHAR(50) DEFAULT 'new',     -- new, contacted, qualified, booked, converted, lost  
  ai_conversation_history JSONB,        -- Full chat log  
  
  -- Compliance  
  consent_given BOOLEAN DEFAULT FALSE,  
  consent_timestamp TIMESTAMPTZ,  
  gdpr_opt_in BOOLEAN DEFAULT FALSE,  
  
  -- Timestamps  
  created_at TIMESTAMPTZ DEFAULT NOW(),  
  updated_at TIMESTAMPTZ DEFAULT NOW(),  
  first_response_at TIMESTAMPTZ,  
  qualified_at TIMESTAMPTZ,  
  booked_at TIMESTAMPTZ  
);  
  
CREATE INDEX idx_leads_client_score ON leads(client_id, lead_score);  
CREATE INDEX idx_leads_created ON leads(created_at DESC);  
CREATE INDEX idx_leads_email ON leads(email);  
CREATE INDEX idx_leads_phone ON leads(phone);
```

5.2 Client Configuration Entity

```
CREATE TABLE client_configs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  client_name VARCHAR(255) NOT NULL,  
  industry VARCHAR(50) CHECK (industry IN ('solar', 'interior_design')),
```

```
-- AI Configuration
ai_prompt_system TEXT,           -- System prompt for this client
ai_temperature DECIMAL(3,2) DEFAULT 0.7,
qualification_questions JSONB,    -- Array of question objects

-- Scoring Rules (JSONB)
scoring_rules JSONB,

-- Integration Settings
crm_type VARCHAR(50),            -- ghl, hubspot, pipedrive
crm_credentials_encrypted TEXT,

-- Communication Preferences
preferred_channels JSONB,        -- ["email", "sms", "whatsapp"]
business_hours JSONB,           -- {"monday": {"open": "09:00", "close": "18:00"}}
timezone VARCHAR(50) DEFAULT 'UTC',

-- Follow-up Configuration
follow_up_sequence JSONB,       -- Customizable per client

created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

6. INTEGRATION MATRIX

6.1 Lead Sources → n8n

Source	Integration Method	Trigger Type	Notes
Website Forms	n8n Webhook Node	HTTP POST	Embed webhook URL in form action
Facebook Lead Ads	Facebook Graph API	Polling (15 min)	Or Zapier bridge initially
Google Ads	Google Ads API	Webhook via Zapier	Leads directly to webhook
WhatsApp	WhatsApp Business API	Webhook	Meta Business verification required
Phone Calls	Twilio + AI Voice	Webhook	Premium feature
Manual Import	CSV Upload + n8n	Manual trigger	Bulk import workflow

6.2 n8n → CRM

CRM	API Method	n8n Node	Data Sync
GoHighLevel	REST API	HTTP Request	Contacts, Opportunities, Custom Fields
HubSpot	REST API	HubSpot Node	Contacts, Deals, Timeline Events
Pipedrive	REST API	Pipedrive Node	Persons, Deals, Activities

6.3 n8n → Communication

Channel	Provider	n8n Node	Template System
Email	SendGrid	SendGrid Node	Dynamic MJML templates
SMS	Twilio	Twilio Node	Per-client message variants
WhatsApp	WhatsApp Business API	HTTP Request	Template messages (Meta approved)

7. AI PROMPT ARCHITECTURE

7.1 System Prompt Structure (Per Industry)

```
You are [Client Name]'s AI Lead Assistant. Your job is to:
1. Respond instantly and warmly to new enquiries
2. Ask 3-4 qualification questions to understand the prospect
3. Score the lead as HOT, WARM, or COLD based on answers
4. Move HOT leads toward booking a consultation

INDUSTRY: Solar Installation
QUALIFICATION QUESTIONS:
1. What is your average monthly electricity bill?
2. Do you own your home? (Critical for solar)
3. Are you looking to install within the next 3 months?
4. What is your preferred contact method?

SCORING RULES:
- HOT: Owns home + bill > $150 + wants install within 3 months
- WARM: Owns home + interested but timeline unclear
- COLD: Renter, or just browsing, or timeline > 6 months

TONE: Professional, helpful, concise. Max 2-3 sentences per message.
LANGUAGE: Respond in the same language as the prospect's message.
```

7.2 Structured Output Format (JSON Mode)

```
{
  "response_text": "Thank you for your interest in solar! To help us prepare the best recommendation,
  ↪ could you share your average monthly electricity bill?",
  "lead_score": "WARM",
```

```
"score_reason": "Prospect owns home but timeline is unclear",
"next_action": "ask_qualification_question_2",
"language_detected": "en",
"sentiment": "positive"
}
```

8. SECURITY & COMPLIANCE ARCHITECTURE

8.1 Data Protection by Region

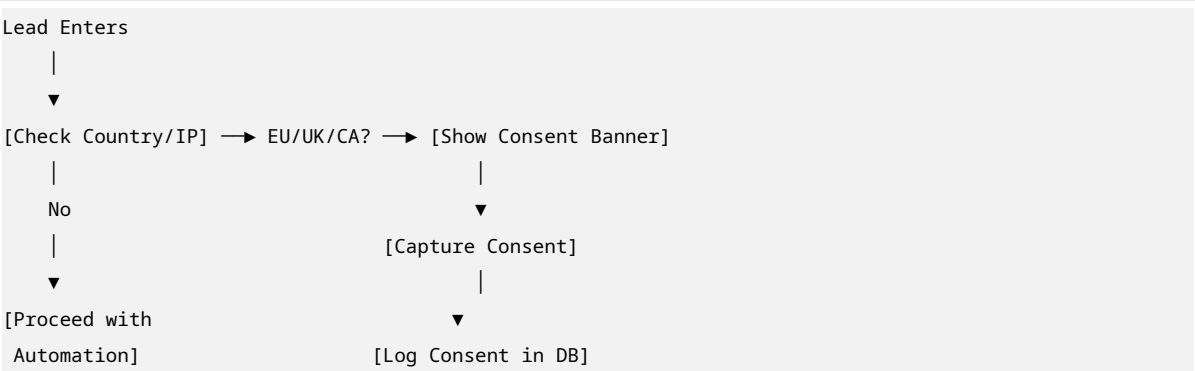
Regulation	Requirement	Implementation
GDPR (EU/UK)	Consent, Right to deletion, Data portability	Consent checkbox pre-capture; automated deletion workflow; data export endpoint
CCPA (California)	Opt-out, Disclosure	Unsubscribe link in all comms; privacy policy automation
CAN-SPAM (US)	Unsubscribe, Physical address	Footer in all emails; one-click unsubscribe
CASL (Canada)	Express consent	Double opt-in for Canadian leads
LGPD (Brazil)	Similar to GDPR	Apply GDPR workflow to Brazilian leads

8.2 Security Measures

SECURITY LAYER

- n8n Basic Auth + API Key Protection
- Webhook URL Signing (HMAC)
- Encrypted Credentials in n8n
- PostgreSQL Row-Level Security
- HTTPS Only (Let's Encrypt)
- Rate Limiting on Webhooks
- IP Whitelisting for Admin Access
- Regular Credential Rotation

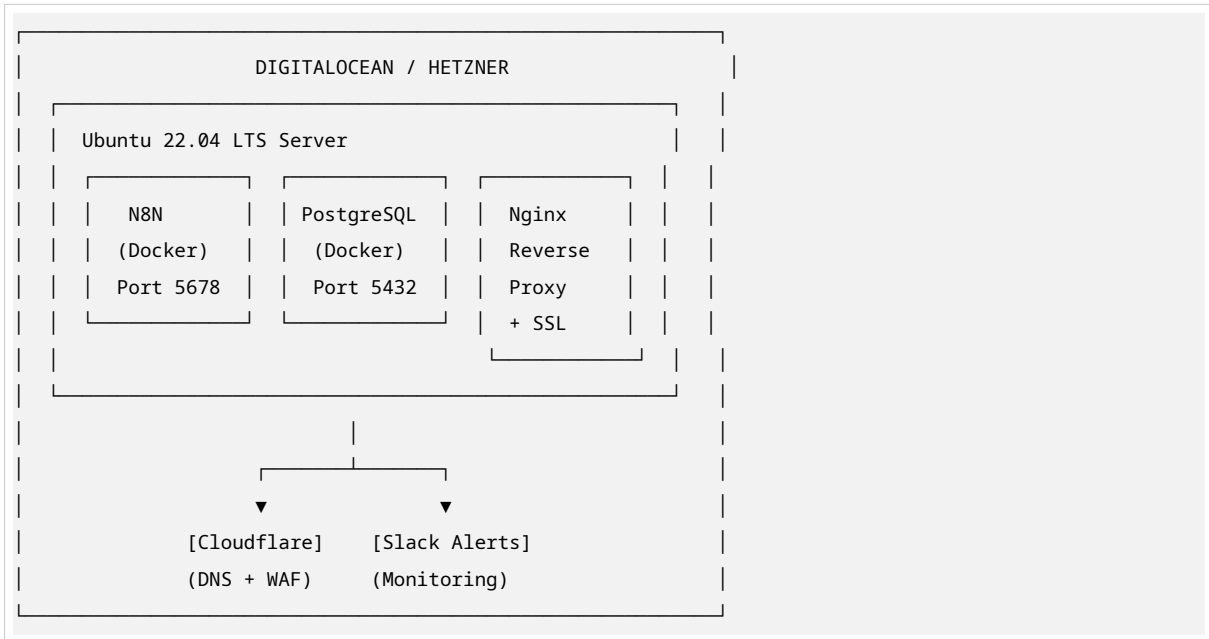
8.3 Consent Workflow



[GDPR Compliant
Processing]

9. DEPLOYMENT ARCHITECTURE

9.1 Infrastructure Setup



9.2 Docker Compose Configuration

```
version: '3.8'
services:
  n8n:
    image: n8nio/n8n:latest
    restart: always
    ports:
      - "127.0.0.1:5678:5678"
    environment:
      - N8N_BASIC_AUTH_ACTIVE=true
      - N8N_BASIC_AUTH_USER=${N8N_USER}
      - N8N_BASIC_AUTH_PASSWORD=${N8N_PASSWORD}
      - DB_TYPE=postgresdb
      - DB_POSTGRESDB_HOST=postgres
      - DB_POSTGRESDB_DATABASE=n8n
      - DB_POSTGRESDB_USER=${DB_USER}
      - DB_POSTGRESDB_PASSWORD=${DB_PASSWORD}
      - WEBHOOK_URL=https://n8n.nexomate.com/
    volumes:
      - ~/.n8n:/home/node/.n8n
    depends_on:
```

```

- postgres

postgres:
  image: postgres:15-alpine
  restart: always
  environment:
    - POSTGRES_USER=${DB_USER}
    - POSTGRES_PASSWORD=${DB_PASSWORD}
    - POSTGRES_DB=nexomate
  volumes:
    - postgres_data:/var/lib/postgresql/data
  ports:
    - "127.0.0.1:5432:5432"

nginx:
  image: nginx:alpine
  restart: always
  ports:
    - "80:80"
    - "443:443"
  volumes:
    - ./nginx.conf:/etc/nginx/nginx.conf
    - ./ssl:/etc/nginx/ssl
  depends_on:
    - n8n

volumes:
  postgres_data:

```

10. MONITORING & OBSERVABILITY

10.1 Error Handling Workflow

```

[Any Workflow Fails]
  |
  ▼
[n8n Error Trigger]
  |
  ▼
[Capture Error Details]
  |
  ▼
[Send Slack Alert]
  Channel: #n8n-alerts
  Content: Workflow Name, Error Message, Timestamp, Lead ID
  |
  ▼
[Log to PostgreSQL]
  Table: error_logs

```

|

▼

[Escalation Rule]

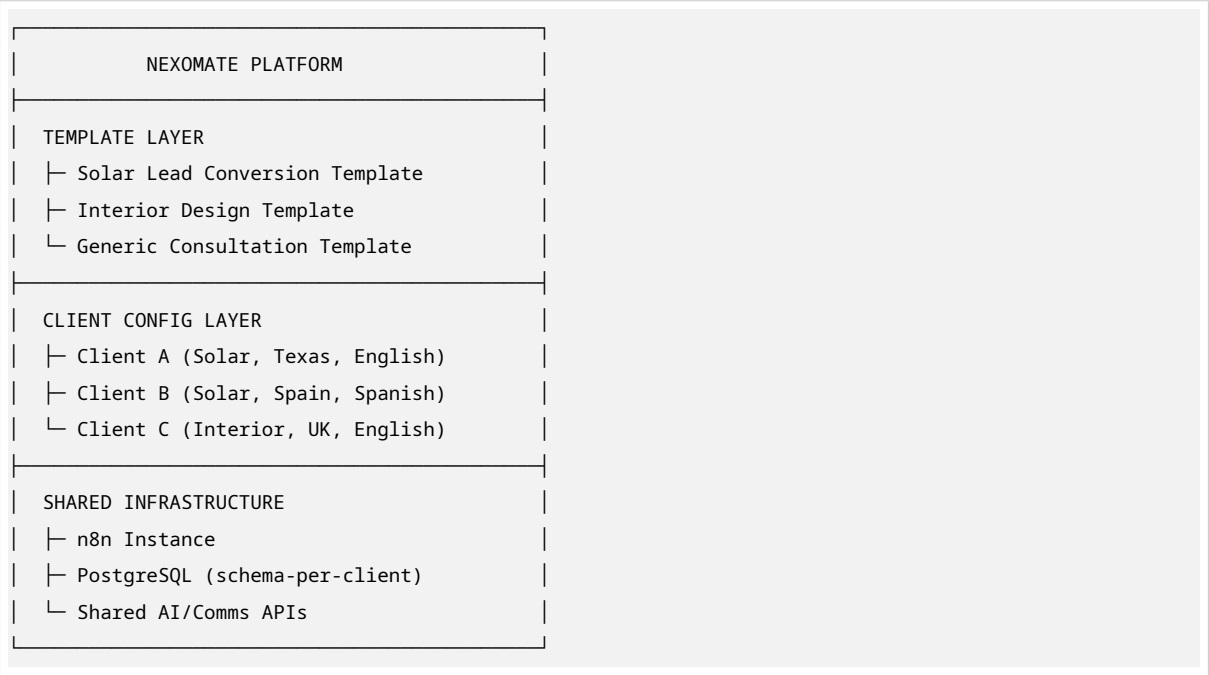
If 3+ errors in 10 min → Page on-call

10.2 Key Metrics Dashboard

Metric	Source	Alert Threshold
Webhook Response Time	n8n Logs	> 5 seconds
AI API Failures	OpenAI/Claude	> 5% error rate
CRM Sync Failures	n8n Error Node	Any failure
Lead Processing Lag	PostgreSQL	> 2 minutes
Daily Lead Volume	PostgreSQL	Drop > 50%
Conversion Rate	CRM	Client-specific

11. MULTI-TENANCY & SCALABILITY

11.1 Template-Based Deployment



11.2 Scaling Path

Phase	Clients	Infrastructure	Cost/Month
MVP	1-3	Single DO droplet (\$24)	~\$50-80
Growth	4-10	Upgraded droplet + managed DB (\$48)	~\$150-200
Scale	10-30	Multiple n8n workers + load balancer (\$120)	~\$400-600

Table 7 – continued

Phase	Clients	Infrastructure	Cost/Month
Enterprise	30+	Kubernetes cluster + dedicated AI infra	Custom

12. COST ESTIMATION (Per Client, Monthly)

Component	Tool	Cost (USD)
n8n Hosting	DigitalOcean	\$12-24 (shared across clients)
AI (OpenAI)	GPT-4o	\$20-50 (depends on lead volume)
Email	SendGrid	\$19.95 (100k emails)
SMS	Twilio	\$0.0075/msg (~\$15-30)
WhatsApp	Meta API	\$0.005-0.05/msg (~\$10-20)
CRM	GoHighLevel	\$97-297 (per client, passed through)
Database	Supabase	\$25 (shared)
Voice AI	Bland AI	\$0.09/min (~\$20-50)
Total Variable		~\$120-250 per client

Pricing Recommendation: \$1,500 setup + \$500-800/month retainer = healthy margin.

13. IMPLEMENTATION ROADMAP

Phase 1: Foundation (Weeks 1-2)

- Set up n8n + PostgreSQL infrastructure
- Build core webhook ingestion pipeline
- Integrate OpenAI for basic response generation
- Connect first CRM (GoHighLevel)
- Build email/SMS notification workflows

Phase 2: Intelligence (Weeks 3-4)

- Implement qualification question engine
- Build scoring algorithm (HOT/WARM/COLD)
- Create follow-up sequence automation
- Add appointment booking integration
- Build error handling + monitoring

Phase 3: Scale (Weeks 5-6)

- Create industry templates (Solar, Interior Design)
- Add multi-language support

- Implement GDPR/compliance workflows
- Build client onboarding automation
- Create internal reporting dashboard

Phase 4: Premium (Weeks 7-8)

- Integrate AI Voice Calling (Bland AI)
- Add WhatsApp Business API
- Build review/referral post-sale automation
- Implement advanced analytics
- Prepare for second market entry

14. RISK MITIGATION

Risk	Impact	Mitigation
n8n workflow failure	High	Error workflows, redundant triggers, manual fallback
AI API rate limiting	Medium	Retry logic, fallback to Claude, queue system
CRM API changes	Medium	Abstraction layer, version pinning
Data breach	Critical	Encryption, least-privilege access, regular audits
Client churn	Medium	Monthly value reports, optimization loops
Founder dependency	High	Document everything, build self-service onboarding

15. DECISION LOG (For Wednesday Discussion)

Decision Needed	Options	Recommendation
Primary CRM	GHL vs HubSpot vs Pipedrive	Start with GHL (solar market), offer HubSpot as enterprise upsell
Hosting	Self-hosted vs n8n Cloud	Self-hosted (cost + control), migrate to cloud if ops overhead grows
AI Model	GPT-4o vs Claude vs Gemini	GPT-4o primary, Claude fallback for complex reasoning
Voice AI	Phase 1 or later	Phase 2 –it’s a premium upsell, not MVP
Multi-language	Day 1 or later	Day 1 for Spanish (US + Spain solar market), others on demand
Pricing Model	Project vs Retainer	Hybrid: \$1,500 setup + \$500/month base + performance tiers
First Market	US vs UK vs Australia	US solar market –highest willingness to pay, English-speaking

16. APPENDIX

A. n8n Webhook Security

```
// HMAC Signature Validation in n8n Function Node
const crypto = require('crypto');
const secret = $env.WEBHOOK_SECRET;
const signature = $input.first().json.headers['x-signature'];
const payload = JSON.stringify($input.first().json.body);

const expected = crypto
  .createHmac('sha256', secret)
  .update(payload)
  .digest('hex');

if (signature !== expected) {
  return [{ json: { error: 'Invalid signature' } }];
}
return $input.all();
```

B. AI Response Caching Strategy

- Cache common responses (business hours, location, basic pricing) for 1 hour
- Reduces AI API costs by ~30%
- Implemented via n8n IF node + PostgreSQL cache table

C. Backup Strategy

- Daily automated PostgreSQL dumps to S3-compatible storage
- n8n workflow exports versioned in Git
- Recovery time objective: < 4 hours

Document Version: 1.0

Prepared By: Technical Lead — Nexomate

Date: August 2026

Next Review: Post-MVP Launch

“Build the system you’d want to maintain at 3 AM when a client’s pipeline breaks.”